

SHETH MVLU COLLEGE

SUBJECT:- AI

Practical 3

AIM:- Naive Bayes' Classifier

- Implement the Naive Bayes' algorithm for classification.
- Train a Naive Bayes' model using a given dataset and calculate class probabilities.
- Evaluate the accuracy of the model on test data and analyze the results.

```
1. Importing the libraries
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

print("Libraries Imported Successfully")

2. Libraries Imported Successfully

3. from google.colab import files
   uploaded = files.upload()

   [Choose File] No file chosen
   Saving bank.csv to bank (2).csv

4. # Importing the dataset
   dataset = pd.read_csv("bank.csv", sep=";")

   print("Dataset Loaded Successfully")

   Dataset Loaded Successfully

5. print(dataset.head())

   age  job  marital  education  default  balance  housing  loan  \
0  30  unemployed  married  primary    no     1787    no     no
1  33  services    married  secondary no     4789    yes    yes
2  35  management single  tertiary no     1358    yes    no
3  38  management married  tertiary no     1476    yes    yes
4  59  blue-collar married  secondary no         0    yes    no

   contact  day month  duration  campaign  pdays  previous  poutcome  y
0  cellular  19  oct     79       1      -1         0  unknown  no
1  cellular  11  may     220       1     339         4  failure  no
```

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The image displays two screenshots of a Jupyter Notebook interface, likely Google Colab, showing data exploration code and output for a dataset. The browser tabs at the top include "Naive Bayes Classifi...", "Untitled document", "AI Prac 6 T111 .jpy", "AI-Prac-2/AI PRAC 2", and "sayedaneescs24254...". The address bar shows a Google Drive link: colab.research.google.com/drive/17puaqWKPVEBdLeiEGKWhgtjFR0d0yNq?usp=sharing.

First Screenshot:

```
print(dataset.head())
```

	age	job	marital	education	default	balance	housing	loan	
0	30	unemployed	married	primary	no	1787	no	no	
1	33	services	married	secondary	no	4789	yes	yes	
2	35	management	single	tertiary	no	1350	yes	no	
3	30	management	married	tertiary	no	1476	yes	yes	
4	59	blue-collar	married	secondary	no	0	yes	no	

```
print(dataset.info())
```

```
print("Dataset Shape :", dataset.shape)
```

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 4521 entries, 0 to 4520

Data columns (total 17 columns):

#	column	Non-Null Count	Dtype
0	age	4521 non-null	int64
1	job	4521 non-null	object
2	marital	4521 non-null	object
3	education	4521 non-null	object
4	default	4521 non-null	object
5	balance	4521 non-null	int64
6	housing	4521 non-null	object
7	loan	4521 non-null	object
8	contact	4521 non-null	object
9	day	4521 non-null	int64
10	month	4521 non-null	object
11	duration	4521 non-null	int64
12	campaign	4521 non-null	int64
13	pdays	4521 non-null	int64
14	previous	4521 non-null	int64
15	poutcome	4521 non-null	int64
16	y	4521 non-null	int64

Dataset Shape : (4521, 17)

Second Screenshot:

```
print(dataset.isnull().sum())
```

	age	job	marital	education	default	balance	housing	loan	contact	day	month	duration	campaign	pdays	previous	poutcome	y	dtype
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	int64

Dataset Shape : (4521, 17)

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The image displays two screenshots of a Google Colab notebook titled 'AI Prac 6 T111 .ipynb'. The notebook is open in a web browser, showing the code editor and the output of the executed cells. The code is written in Python and uses the sklearn library for data preprocessing and model training.

Code Snippet 1 (Top Screenshot):

```
from sklearn.preprocessing import LabelEncoder
encoder = LabelEncoder()
for column in dataset.columns:
    if dataset[column].dtype == 'object':
        dataset[column] = encoder.fit_transform(dataset[column])
print("Categorical Data Encoded Successfully")
```

Code Snippet 2 (Bottom Screenshot):

```
# Independent Variable
X = dataset.iloc[:, :-1].values
# Dependent Variable
y = dataset.iloc[:, -1].values
print("X Shape :", X.shape)
print("Y Shape :", y.shape)
X Shape : (4521, 16)
Y Shape : (4521,)

from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=0
)
print("Training and Testing Data Created")
```

The output of the first cell shows 'Categorical Data Encoded Successfully'. The output of the second cell shows the shapes of the independent variable (X) and dependent variable (y), and the message 'Training and Testing Data Created'.

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